

Metabolic Changes in Blood and Liver During Development and Early Treatment of Experimental Fatty Liver and Ketosis in Cows¹

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ABSTRACT

Eighteen cows were assigned in equal numbers to three groups: control, ketosis induction by using feed restriction plus dietary 1,3-butanediol to provide ketone bodies, and glucose treatment with 484 g/d of glucose infused intraduodenally starting 7 d after beginning ketosis induction. Ketosis induction, begun at d 15 postpartum, caused ketonemia and gradual development of clinical ketosis by d 40 to 45. None of the cows in the control or glucose-treated groups became ketotic. Concentrations of NEFA in plasma of cows that became ketotic increased 3.0-, 2.6-, and 1.9-fold at 3 wk before, 2 wk before, and at ketosis, respectively, but increased nonsignificantly for glucose-treated cows. Concurrently, β -hydroxybutyrate increased 3.5-, 5.8- and 8.4-fold for cows that became ketotic but 1.6-fold or less for glucose-treated cows. Plasma acetate increased dramatically 2 wk before ketosis. Liver

glycogen content decreased to nearly 0 by 2 wk before ketosis occurred, but it increased to prepartal values in glucose-treated cows. Liver triglycerides averaged 2.0% of wet weight at d 5 for all cows but increased to 8 to 10% for about 2 wk before ketosis occurred. Microscopy of liver samples demonstrated progressive accumulation of lipid globules, which began in hepatocytes near the central vein and progressed toward the portal triad. Visible lipid content reached a peak 2 wk before ketosis. Hepatic *in vitro* gluconeogenic capacity decreased significantly for ketosis induction protocol cows when clinical ketosis was detected. Results indicate that experimental ketosis was preceded by metabolic abnormalities up to 2 wk before clinical ketosis occurred. The key events for onset of clinical ketosis, however, were not elucidated.

(Key words: ketosis, fatty liver, dairy cows)

Abbreviation key: BD = 1,3-butanediol; BHBA = β -hydroxybutyric acid; FRBD = feed restriction coupled with BD; GT = glucose treatment during ketosis induction treatment; KIP = ketosis induction protocol treatment.

INTRODUCTION

The characteristics of clinical cases of bovine lactation ketosis have been known for many years. These include observable signs such as hypophagia, decreased milk production, rapid loss of BW, lethargy or hyperexcitability (28), and metabolic conditions such as hyperketonemia, hypoglycemia, hypoinsulinemia, decreased hepatic glycogen, increased hepatic triglycerides, and increased NEFA in plasma (1). Subclinical ketosis also

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is common in high producing dairy cows and encompasses mild hyperketonemia, some loss of milk production, and lower concentrations of blood glucose (1).

It is generally accepted that the ketotic syndrome is related to high energy requirement for milk production, which often cannot be met in high producing cows by the normal intake of energy during early lactation (1, 15). Little is known, however, about the metabolic progression from subclinical to clinical states. The alterations of metabolic control mechanisms admittedly are complicated (14), but fatty liver has been implicated in the process. Recent work from Finland showed that the incidence of ketosis was 30% in cows with fatty liver but only 10% for those without fatty liver (13). Inability to detect metabolic abnormalities before signs of clinical ketosis become evident in cows has precluded in-depth studies of the progressive development of spontaneous, on-farm ketosis.

Attempts to induce ketosis experimentally by starvation have resulted in metabolic profiles similar to those of spontaneously ketotic cows (2, 7) but have been inadequate to evaluate the progressive development of ketosis. Inducing ketosis in early lactation cows by using moderate feed restriction coupled with feeding 1,3-butanediol (FRBD) has been successful, however (24, 25). The 1,3-butanediol (BD) serves as a ketone body precursor that is absorbed across the rumen epithelium. Starting the induction protocol at d 14 postpartum has resulted in cows gradually developing ketosis over about a 21- to 28-d period.

The purpose of our study was to use a ketosis induction protocol (KIP) (24, 25) to examine the developmental pattern of metabolic changes associated with ketosis. Production indices, metabolites and insulin in blood, composition of liver, and hepatic *in vitro* gluconeogenic capacity were measured at nearly weekly intervals during development of ketosis to accomplish this objective. Another objective was to test the effects of a continuous infusion of glucose into the duodenum during the KIP. The sampling schedule of the experiment was designed to evaluate whether specific metabolic changes occur at the transition between subclinical and clinical ketosis.

MATERIALS AND METHODS

Design of Experiment and Management of Cows

One Brown Swiss and 17 Holstein cows, averaging 595 kg of BW at 4 d postpartum, were used in a completely randomized design. Six cows initially were assigned to each of three treatment groups: control, KIP, or glucose treatment during ketosis induction (GT). One Holstein cow did not respond to the KIP, and data from that cow were omitted. Another KIP cow was injured after developing severe ketosis and was removed from the study. Thus, the data in this report are based on results from 16 cows.

For about 30 d before parturition, all cows were housed and fed hay for *ad libitum* intake in a lot for dry, pregnant cows at the Iowa State University dairy farm. After d 4 postpartum, 14 of the cows were housed in a tie-stall barn with other herd cows. The other four cows were housed in a separate metabolism facility (for measurements not included in this study) but otherwise were handled identically to the 14 cows. All cows on the study were fed silage and grain mixture (approximately 60% forage and 40% concentrate) for *ad libitum* intake plus long-stemmed alfalfa hay. Feed ingredients, except hay, are listed in Table 1. Fresh feed was given twice daily between 0900 and 1000 h and between 1600 and 1800 h. Orts were collected daily to determine intake of silage and grain mixture. The long-stemmed alfalfa hay was fed to an approximate total of 6 kg/d for each cow. Control treatment consisted of continuing *ad libitum* feeding of the silage and grain mixture plus the 6 kg of hay throughout the experiment.

The KIP treatment consisted of the following: cows gradually were induced into clinical ketosis by using the 20% feed restriction plus dietary FRBD protocol published earlier (24), except cows were not overfed during the dry period. Specifically, at d 14 postpartum, intake of silage plus grain mix was restricted to 80% of the average intake of the previous 5 d, and the ketone body precursor, BD, was added to the feed at 4% of DMI. (The BD, i.e., 1,3-butylene glycol, was donated by the Celanese Chemical Co., Dallas, TX.) At d 18 postpartum, BD supplementation was increased to 8% of DMI. The FRBD continued until ketosis was detected. Supplementation of BD was held at a constant 8% of DMI after d

TABLE 1. Ingredients of the mixed diet fed to cows.¹

Ingredient	Silage and grain mixture (% of DM)
Silage mixture	
Alfalfa haylage	33.3
Corn silage	20.6
Corn gluten feed	6.3
Total	60.2
Grain mixture	
Corn grain	32.2
Soybean oil meal	6.6
Vitamin and mineral supplement ²	1.0
Total	39.8
Diet formulated to be	
CP, %	17.4
ADF, %	20.5
NE _L , Mcal/kg	1.71

¹Cows also were fed long-stemmed alfalfa hay at approximately 6 kg/d.

²The vitamin and mineral supplement, on a dietary DM basis, was a mixture of .1% sodium bicarbonate, .1% magnesium oxide, .3% dicalcium phosphate, .3% calcium carbonate, .2% trace-mineralized salt, and .05% vitamins A, D, E, and K premix. The trace-mineralized salt contained not more than 98.0% and not less than 92.0% NaCl, .35% Zn, .28% Mn, .175% Fe, .035% Cu, .007% I, and .007% Co. The vitamin premix contained 4,400,000 IU/kg of vitamin A (vitamin A acetate); 4,400,000 IU/kg of vitamin D₃; 2200 IU/kg of vitamin E (dl- α -tocopherol acetate); and 1500 mg/kg of vitamin K (menadione dimethylpyrimidinol bisulfite).

18 postpartum, which is in contrast with earlier work with FRBD in which variable amounts of BD were fed (24). An average of 900 g of BD/d was consumed daily, thereby resulting in an estimated 14.7% of the metabolizable energy of the FRBD diet originating from BD.

Cows in the GT group were treated identically to those in the KIP group except for the intraduodenal glucose infusion. At d 23 of lactation, a small polyvinyl catheter was placed in the proximal duodenum about 20 cm caudal to the pylorus. Surgery was performed with cows standing upright. Xylocaine® (2% lidocaine hydrochloride, Med Tech, Inc., Elkwood, KS) was injected along the right rib cage from the 9th to the 13th rib. An 18-cm incision was made at the 11th intracostal space in the region of the costochondral junction. Blunt dissection was used to penetrate the peritoneal cavity, and the duodenum was externalized temporarily. An indwelling catheter was placed into the duodenum, sutured to the duodenal surface by

using an absorbent sponge for support, and externalized at the 12th intercostal space. The incision was closed with surgical staples. Cows decreased feed intake for one feeding after surgery but returned to normal by the next morning. Milk production was not affected, and inflammation at the incision site was minimal. At d 25 of lactation, tygon tubing was connected to the catheter, and a peristaltic pump (model TM 10, Sigmamotor, Middleport, NY) was used to infuse a glucose solution (.35 g/ml) at $.96 \pm .03$ ml/min (484 g/d of glucose) for 15 d. The intraduodenal infusion of glucose was intended to coincide with the time when subclinical and clinical ketosis would be developing for KIP. The FRBD protocol was continued for 7 d after glucose infusion was terminated on d 39 postpartum. One infusion of glucose was terminated 7 d after it began because of hyperexcitability and nervousness of the cow, but FRBD was continued. Two other cows showed extreme hyperexcitability during the last 3 d of glucose infusion.

Determination of the onset of clinical ketosis was by the Iowa State University Veterinary Service and was based upon several factors (12), including the presence of high concentrations of ketone bodies in urine (KETO-DIASTIX, Ames Division, Miles Laboratories, Inc., Elkhart, IN), altered behavior indicative of ketosis (28), decreased milk production, and voluntary decreases in the already restricted consumption of the FRBD diet. When a cow was diagnosed as clinically ketotic, she was returned to ad libitum intake without BD. No further treatment for ketosis was planned, but two cows required treatment with 500 ml of 50% dextrose (Travenol Laboratories, Inc., Deerfield, IL) i.v. when severe clinical ketosis was detected. No GT or control cows became clinically ketotic.

Sampling and Analytical Procedures

Milk production was recorded daily, and cows were weighed on alternate days. Sampling of blood and liver was scheduled for about d 14 prepartum, d 5 postpartum, d 13 postpartum just before start of FRBD, and then weekly thereafter until ketosis was detected, followed by a sample 2 wk later. The sampling schedule was complicated by the unpredicta-

TABLE 2. Summary of sampling of blood and liver for cows on study.

Schedule sampling time ¹	Average day of sampling		Maximum number of cows contributing to analysis for each treatment		
	$\bar{X}$	SD	Control	KIP	GT
About 2 wk prepartum	-11.4	1.5	6	4	6
Between 4 and 7 d postpartum	5.4	.4	6	4	6
Just before start of FRBD	13.2	.3	6	4	6
About 1 wk after start of FRBD	20.8	.3	6	4	6
About 2 wk after start of FRBD	28.1	.3	4	3	6
About 3 wk after start of FRBD	34.9	.4	6	3	5
At clinical ketosis diagnosis	44.2	5.8	6	4	6
About 2 wk after clinical ketosis	60.1	3.7	5	4	5

¹Scheduled sampling times are presented in reference to KIP treatment. Timing was similar for control and GT cows. KIP = Ketosis induction protocol treatment, GT = glucose-treated during induction treatment, FRBD = feed restriction, 1,3-butanediol.

bility of development of ketosis in KIP cows and concern for cow health during repeated liver biopsies. Table 2 shows the mean sampling day for each time period and number of cows contributing to each mean. Grouping of control cow samples was done to correspond to the days of sampling for KIP cows. The standard deviation for all sampling times except those at ketosis were low; therefore, mean sampling day for each period will be used for presenting results.

Clinical ketosis was diagnosed at a mean of 44 d postpartum for KIP cows (Table 2). The range was 24 to 56 d; the diagnosis at d 24 was considerably earlier than the others. When ketosis was detected, the data points from that time were analyzed in the ketosis grouping, and any scheduled preketotic samples not taken were considered missing. Similarly, the data from one GT cow for which glucose infusion was terminated early were treated as missing for the 2nd scheduled wk of infusion.

Blood samples were collected into heparinized tubes that contained 150 μ l of 4% NaF to inhibit glycolysis. Plasma was prepared, stored at -20°C, and assayed for concentrations of glucose (glucose Trinder 500 kit, Sigma Chemical Co., St. Louis, MO), NEFA (NEFA C kit, WAKO Chemicals, Dallas, TX), and acetate (21). Protein-free filtrates of plasma (30) were prepared and assayed for β -hydroxybutyric acid (BHBA) (31).

Concentrations of insulin in blood plasma were determined by using standard double-antibody radioimmunoassay (11). Antiserum to bovine insulin (ICN Immunobiologicals, Lisle,

IL) was used as the primary antibody, and goat anti-guinea pig IgG (BioTek Research, Inc., Lenexa, KS) was used as the second antibody. Intraassay and interassay coefficients of variation were 7 and 3%, respectively. Immediately after blood samplings, 3 to 4 g of liver were taken by puncture biopsy (18). Liver biopsy did not result in decreased feed intake or milk production for most cows. Some cows, however, decreased their a.m. feed intake but resumed normal intake by the p.m. feeding. Approximately 1 g was frozen immediately at -20°C for later determination of glycogen and triglyceride content. Approximately .2 g of liver was fixed in 10% neutral buffered formalin. The remaining samples of liver were placed in cold .15 M NaCl buffered with phosphate (.015 M) to pH 7.4, stored on ice, and transported immediately to the laboratory. In vitro incubations were begun within 60 min after biopsy by procedures already described (24), with the exception that incubations were conducted with 2 mM Ca in the media. Sodium-[2-¹⁴C]propionate was used as the substrate for estimation of in vitro gluconeogenic capacity. Glucose and CO₂ were recovered from media after incubation (24). The effect of omitting Ca⁺⁺ from incubation media also was tested (data not reported). Five- to sixfold more transfer of ¹⁴C into propionate or CO₂ was observed when Ca⁺⁺ was present. Relative differences between treatments, however, were similar.

Frozen liver samples later were thawed for removal from sample vials, refrozen in liquid N, and pulverized by grinding with a mortar

and pestle. About half of each sample was weighed and homogenized in 5 vol by weight of 10% perchloric acid. The resulting suspension was used for determination of glycogen (19).

The other half of each sample was placed in 10 ml of 2:1 chloroform and methanol. Samples were allowed to shake for at least 2 h, 4 ml of H₂O were added, and samples were vortexed vigorously and then centrifuged. After centrifugation, the methanol-water layer was aspirated, liver tissue was removed, and the remaining chloroform was evaporated under N. The lipid extract then was dissolved in 95% ethanol and assayed for triglycerides (triglyceride GPO Trinder kit, Sigma Chemical Co., St. Louis, MO).

For light microscopy, sections of liver that had been fixed in 10% neutral buffered formalin were embedded in paraffin, sectioned at 5 to 6 μ , and stained with hematoxylin and eosin. Cryostat sections were prepared from adjacent blocks of liver and stained with Oil Red O (Sigma Chemical Co.).

Statistical Analysis

Data were analyzed by using general linear models to compare period (day) differences within treatments and to compare treatment differences at each stage of the experiment (27). A split-plot design was used with treatments as main plots and periods as subplots. Analysis of covariance was attempted by using pretreatment values as covariates, but reduced degrees of freedom and missing values resulted in an unsolvable matrix. Therefore, treatments were tested by using the error term of cow within treatment without covariate, and stage and treatment by stage interactions were tested by using residual error. Least squares means were calculated (27), and significance was declared at $P < .05$ unless otherwise noted.

RESULTS

Milk Production, DMI, and BW Change

Averages for milk production, DMI of mixed diet, and BW change are shown in Figure 1 for each treatment at each stage of the experiment. Milk production (Figure 1A) in-

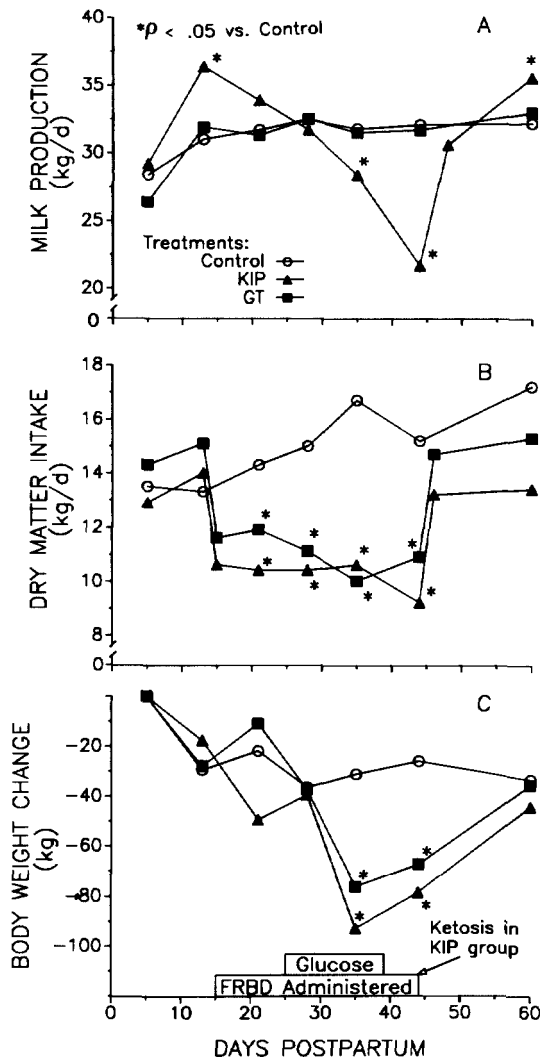


Figure 1. Milk production (A), DMI of mixed diet (B), and BW change (C) for control cows, cows given the ketosis induction protocol (KIP), and cows given glucose treatment (GT). Asterisks refer to differences ($P < .05$) for a particular day for KIP or GT versus control, and FRBD indicates treatment with feed restriction and 1,3-butanediol. Average standard error for control, KIP, and GT groups were A) 1.00, 1.21, and 1.05; B) .8, 1.0, and 1.8; and C) 15.2, 21.4, and 14.2.

creased for all groups between d 5 and 13 postpartum; differences between d 5 and 13 were significant within KIP and GT groups and within controls ($P < .08$). Production of milk for both control and GT cows remained relatively constant for the remainder of the

experiment. The KIP cows gradually decreased in production during treatment, and production became significantly lower than controls at d 35. A further decrease in milk production occurred at onset of ketosis. This result suggests that the FRBD protocol given to cows soon after parturition can establish milk production responses similar to those of on-farm ketosis. Milk production in KIP cows at d 35 was significantly less than at d 13 and 21 but was not less than at d 28. The average decrease in milk production from d 13 to 35 was less than .5 kg/d. This gradual decrease illustrates the difficulty in detecting subclinical ketosis by monitoring milk production. Milk production for KIP cows after recovery (d 60) was significantly higher than for controls.

Intake of DM of the mixed diet (Figure 1B) for KIP and GT cows reflects the feed restriction from d 14 through the ketotic stage at d 44. The intake of long-stemmed hay is not included in Figure 1B. The initial decrease of feed intake for both KIP and GT cows represented approximately 6 Mcal of NE_L . The GT cows consistently ate all feed offered while receiving FRBD, but KIP cows tended to decrease intake further at onset of ketosis ($P < .15$); three of the four KIP cows became anorexic.

Body weights at d 5 postpartum were not different between groups (data not shown), and body condition scores were not taken. Body weight changes (Figure 1C) indicate that endogenous nutrient stores were being used for energy in all three groups of cows, and the rates of loss of BW were similar through d 28. From d 28 through d 35 and 44, however, KIP and GT cows lost BW rapidly. Even though GT cows lost weight, milk production was maintained (Figure 1A), and they showed no outward signs of ketosis. The 2-wk recovery period, marked by a return to normal feed intake (Figure 1B), was sufficient to restore BW to control values for both KIP and GT cows (Figure 1C). The BW losses observed in KIP and GT cows were more variable and not as severe as those in an earlier study with cows overfed in the dry period (24).

Concentrations of Compounds in Blood

Concentrations of glucose in plasma were the same for all three groups at the prepartum

stage (Figure 2A), but they began to diverge at d 5 and 13 postpartum, even though all cows were fed identically until d 14. The trend for lower glucose concentrations for KIP cows at d 13 corresponded with significantly greater milk production for KIP cows (Figure 1A). Glucose concentrations were more variable after d 13 for control and KIP treatments than for the GT cows. In general, glucose concentration decreased for the first 35 d after calving in control cows but decreased significantly from prepartum only at d 28 and 35. The significant decreases correspond to periods of greatest loss of BW (Figure 1C) and possibly reflect when cows were in the most negative energy balance. Glucose concentration in the KIP cows decreased significantly from d 13 and remained low at d 28 and 44. The increase at d 35 is unexplainable, but two KIP cows had unusually high blood glucose concentrations. Changes in concentration of glucose in KIP cows did not parallel the drop in milk production accompanying ketosis; rather, glucose concentration decreased early in ketosis induction. Similar results were observed earlier (24). After recovery, plasma glucose concentrations of KIP cows returned to those observed at d 5 and 13 but tended to be lower than those for controls ($P < .14$).

Glucose concentrations in GT cows for d 5 and 13, even before any treatment began, were unexplainably higher than for KIP cows but were not significantly higher than for controls. Whether this higher concentration of glucose affected the response of GT cows is unknown. Intraduodenal glucose infusions, begun at d 25 postpartum, resulted in increases in plasma glucose over controls at d 35, and the increased glucose concentrations persisted until d 44 even though glucose infusions were terminated at d 39. Two GT cows developed severe nervousness and hyperexcitability during the last week of glucose infusion. Those two cows had glucose concentrations greater than 90 mg/dl. Results from sheep (23) suggest that i.v. infusions of glucose increase plasma insulin concentrations; indeed, further analysis of blood from GT cows showed an increase in insulin concentration (Figure 2B). This suggests that glucose infusions in GT cows were sufficient to increase glucose concentration despite increased insulin or that GT cows were resistant to insulin.

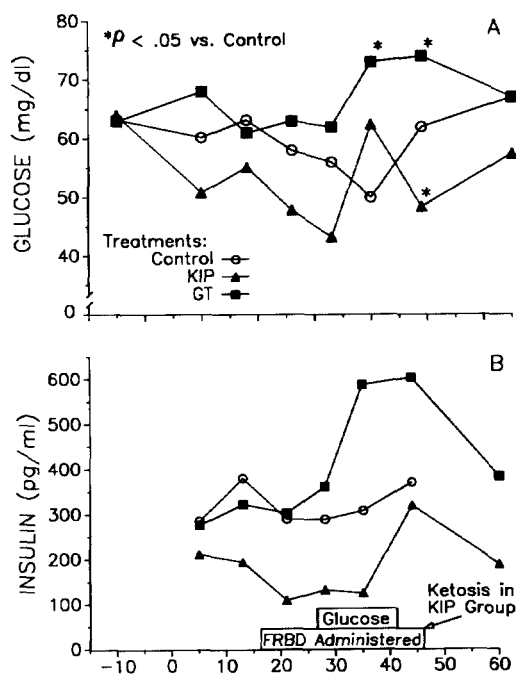


Figure 2. Concentrations in blood plasma of glucose (A) and insulin (B) for control cows, cows given the ketosis induction protocol (KIP), and cows given glucose treatment (GT). Asterisks refer to differences ($P < .05$) for a particular day for KIP or GT versus control, and FRBD indicates treatment with feed restriction and 1,3-butanediol. Average standard error for control, KIP, and GT groups were A) 2.1, 3.6, and 3.7; and B) 81, 101, and 80.

Concentrations of plasma insulin are shown in Figure 2B. Missing values for control cows at d 60 prevented calculation of a least squares mean, and this point is omitted. Insulin concentration for control cows changed very little (Figure 2B), but insulin concentration at d 5 tended to be lower for the KIP group than for controls. Insulin concentration decreased further during the first 2 wk of FRBD. Although values for the KIP were not different from the control group ($P < .20$), insulin concentration for the KIP group was about 40% lower than d 5 at d 21, 28, and 35. Insulin concentration in KIP cows increased at d 44 when clinical ketosis was detected. Insulin concentration for GT cows did not change during the first 2 wk of FRBD, but infusion of glucose resulted in dramatic increases in serum insulin. Means for d 35 and 44 were different from d 5, 13, 21, and 28 within the GT group but were different

from the KIP group only, not the controls. The lack of significant differences from controls presumably is due to the relatively high SE (approximately 80 to 100 pg/ml).

Concentrations of NEFA in plasma, as for glucose, were almost identical prepartum but diverged after parturition before any treatments were begun (Figure 3A). Reasons for the divergence are unknown. Concentrations of NEFA give an indication of the interaction between rates of lipolysis from adipose tissue and uptake of NEFA by liver and other tissues. Concentrations as high as 1.2 mM in plasma have been reported for ketotic cows (4, 29), and plasma NEFA concentrations generally are related inversely to carbohydrate sufficiency in cows. Although NEFA concentrations for control and GT cows were relatively constant during the first 40 d postpartum, for KIP cows, they increased significantly by wk 1 of ketosis induction (d 21) and remained elevated through onset of ketosis at d 44, reflecting mobilization of body reserves after a decrease of caloric intake. Concentrations of NEFA for KIP cows at d 21 and 28 were significantly different from those at d 13. Concentrations of NEFA for KIP cows had increased at d 13 postpartum, but, combined with a low concentration of plasma glucose, this increase suggests that lipid was being mobilized to maintain high milk production (Figure 1A). Concentrations of NEFA for GT cows reflect the trends shown for milk production and plasma glucose concentrations.

Concentrations of BHBA in plasma (Figure 3B) were similar for all three groups from prepartum until treatments were started for KIP and GT groups after d 13 postpartum. Concentration of BHBA was constant throughout for the control group and showed only a slight tendency to increase during administration of FRBD for the GT group. For the KIP group, BHBA concentrations at d 13, 21, 28, and 35 progressively increased and were significantly different from each other. At d 44, BHBA concentrations were different from those at d 21 and 60. Based upon limited sampling, our laboratory reported a pattern of ketogenesis in liver slices similar to that of plasma changes noted for BHBA with KIP cows (24). Our results suggest that, even though NEFA concentration increases and peaks early, BHBA concentration in plasma increases continuously, peaking about 1 wk before signs of clinical ketosis are detected.

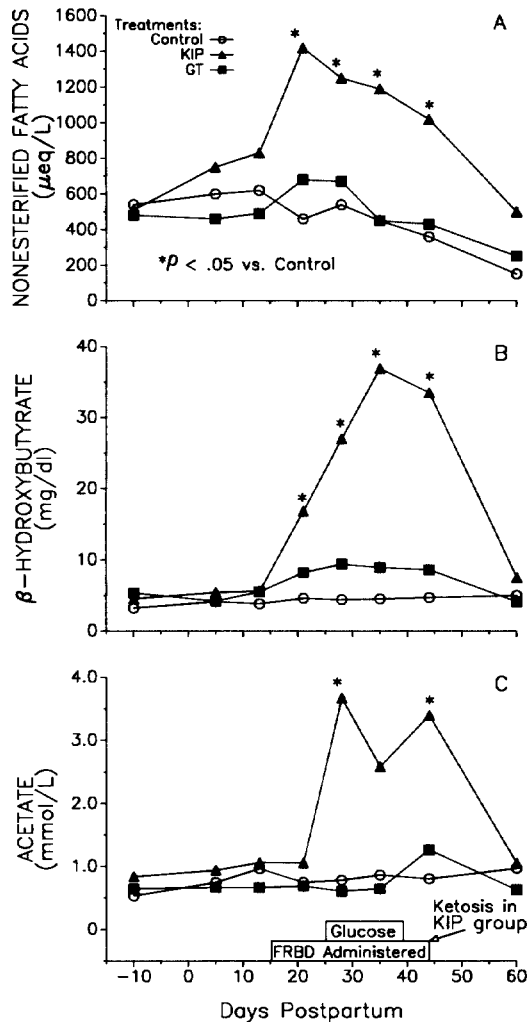


Figure 3. Concentrations in blood plasma of NEFA (A), β -hydroxybutyrate (B), and acetate (C) for control cows, cows given the ketosis induction protocol (KIP), and cows given glucose treatment (GT). Asterisks refer to differences ($P < .05$) for a particular day for KIP or GT versus control, and FRBD indicates treatment with feed restriction and 1,3-butanediol. Average standard error for control, KIP, and GT groups were A) .10, .17, and .11; B) 5.7, 4.1, and 2.2; and C) .10, .49, and .15.

Plasma acetate concentrations (Figure 3C) essentially were the same for all three groups at the prepartum stage and remained at that level throughout the experiment for both the control and GT groups. Acetate concentration increased significantly for the KIP group at d 28 and remained high until ketosis was detect-

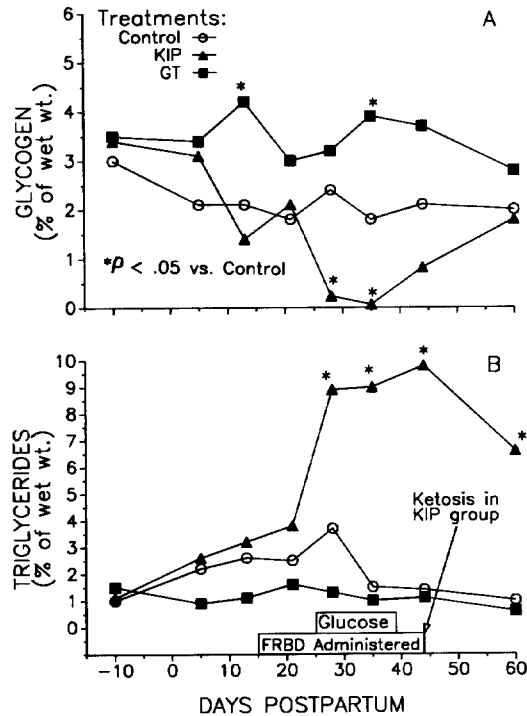


Figure 4. Concentrations [percentage of wet weight (wt.)] of glycogen (A) and triglycerides (B) in liver of control cows, cows given the ketosis induction protocol (KIP), and cows given the glucose treatment (GT). Asterisks refer to differences ($P < .05$) for a particular day for KIP or GT versus control, and FRBD indicates treatment with feed restriction and 1,3-butanediol. Average standard error for control, KIP, and GT groups were A) .47, .86, and .46; and B) .56, 1.00, and .33.

ed. This increase could indicate either production of acetate by the liver in response to increased ketogenesis or decreased peripheral utilization of acetate because of lower insulin concentrations. Acetate production is an alternative to ketogenesis to provide an end product for incomplete metabolism of acetyl coenzyme A. The delay in response of acetate compared with BHBA supports this possibility for KIP cows. The increase in acetate concentration for KIP cows is in contrast with decreases shown for fasting ketosis caused by decreased entry of acetate from the rumen (22).

Liver Composition

Results for liver glycogen and triglycerides are shown in Figure 4. Prepartum concentra-

tions of glycogen (Figure 4A) were somewhat less for control than for GT and KIP groups. By d 5, the divergence had increased, and, by d 13, there were significant differences. During this time, all groups were treated identically, although milk production was greater for KIP cows. For control cows, glycogen decreased slightly after parturition and remained relatively stable thereafter (Figure 4A). Glycogen content of liver decreased from d 5 to 13 for KIP cows, but it increased for GT cows. For KIP cows, decreased glycogen content of liver probably reflects the severity of negative energy balance. Hepatic glycogen content of the KIP cows decreased significantly at d 28, reached nearly zero by d 35, and then increased slightly by the time ketosis was diagnosed at d 44. The direction of changes in glycogen for KIP cows was essentially the opposite of changes in liver triglycerides (Figure 4B). Glycogen content of liver of GT cows decreased slightly ($P < .11$) at the beginning of FRBD and then returned to pretreatment values during glucose infusion.

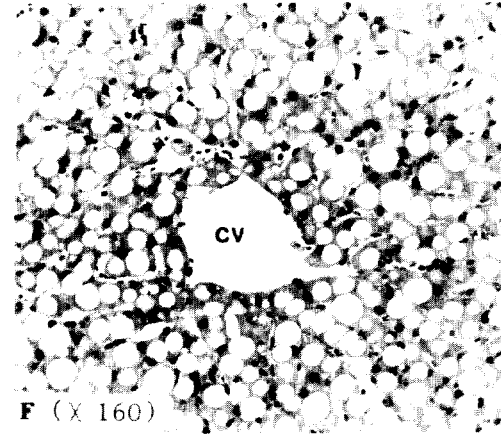
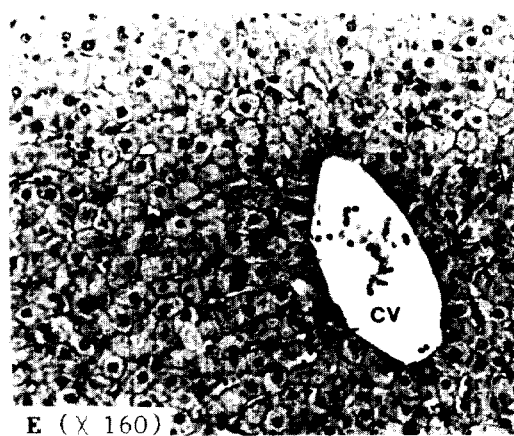
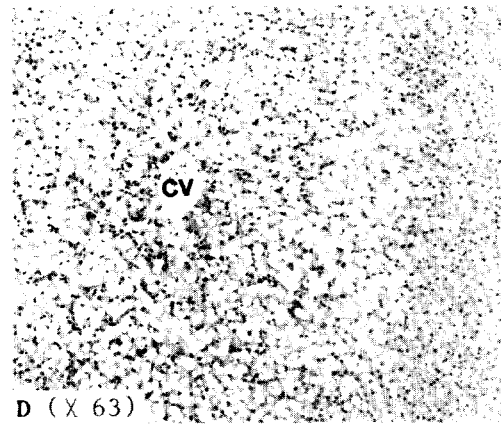
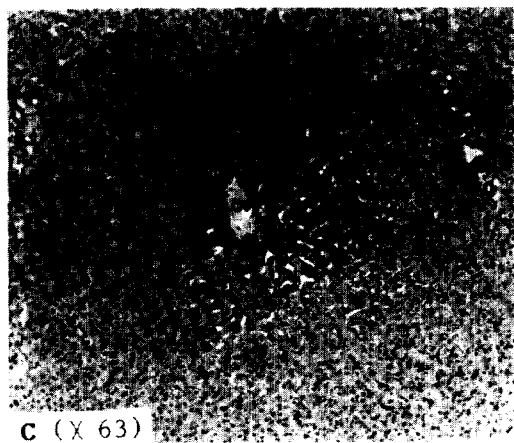
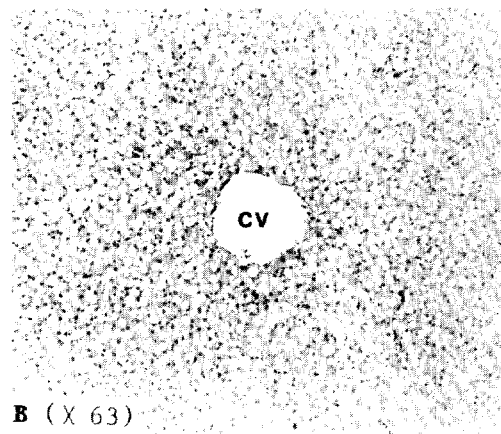
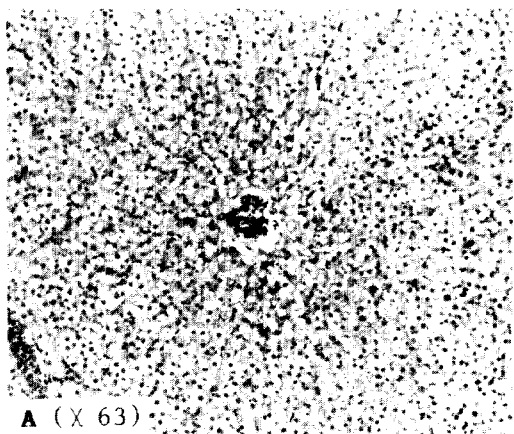
Triglyceride concentrations in liver (Figure 4B) were the same for all three groups at the prepartum stage, but they diverged (not significantly) by d 5 and 13 before any treatments began. Triglycerides increased significantly for the KIP group by d 28 and remained nearly steady through onset of ketosis at d 44, averaging about 10% of liver wet weight. Other studies with ketotic cows also have shown liver triglycerides to 10% of wet weight (3, 16). Unlike other metabolites, liver triglycerides did not return to control or prepartum values after a 2-wk recovery period, indicating a continuing limited ability to transport fat out of the liver. Liver triglycerides of GT cows remained low throughout the experiment.

By using light microscopy (Figure 5), we observed lipid globules in hepatocytes near the

central vein during late gestation and early postpartum stages. Cows in the KIP group continued to accumulate hepatic lipid until KIP began. During the KIP treatment, progressively more hepatocytes between central veins and portal triads had pale and vacuolated cytoplasm. Sections stained with Oil Red O demonstrated that cytoplasmic vacuoles contained neutral lipids, i.e., triglyceride. Extensive hepatocellular vacuolation persisted into the ketosis period and gradually decreased after FRBD was stopped. During recovery, visible lipid was mobilized first from periportal hepatocytes, then from those in midzonal areas, and, finally, from those near central veins.

Figure 5A shows a liver sample from cow 8983 at d 14 postpartum, just before the KIP was begun. The presence of some hepatic lipid is seen in vacuolated cells near the central vein. The progressive accumulation of visible lipid in the livers of KIP cows is illustrated by photographs in Figure 5, B, D, and F. These biopsies from cow 8849 were collected at d 21, 28, and 44 postpartum, respectively. At d 21 (Figure 5B), numerous lipid globules occurred in hepatocytes around the central vein. These globules progressively increased in number and size toward the midzonal and periportal areas (Figure 5, D and F). Hepatocytes were swollen with pale, vacuolated cytoplasm. In later stages of FRBD induction, lipid globules often compressed the cell nucleus to the periphery of the cell. Visible lipid reached its peak 1 to 2 wk before clinical ketosis developed, persisted at that level during the clinical illness, and only decreased slightly for the following 2 wk. This finding illustrates that accumulated lipid disappeared very slowly, even when cows were returned to normal feeding and management and were producing amounts of milk equivalent to controls.

Figure 5. Light microscopy of liver samples from cows 8983 and 8849 on the ketosis induction protocol (KIP) and from control cow 8746. All samples were stained with hematoxylin and eosin. A) Cow 8983, 14 d postpartum. Note the presence of a few vacuolated (lipid laden) hepatocytes chiefly in the zone near the central vein (CV) on the day before KIP began. Lipid droplets were removed during processing. B) Cow 8849, 21 d postpartum. Note the increase in vacuolated hepatocytes after 7 d on the KIP diet. C) Cow 8746, 28 d postpartum, and fed the control diet. Note the paucity of hepatocellular lipid accumulation. D) Cow 8849, 28 d postpartum. Note the numerous vacuolated hepatocytes near the CV and intermediate areas and fewer vacuolated hepatocytes after 14 d on KIP. E) Cow 8746, 44 d postpartum, and fed the control diet. Hepatocytes have pale granular cytoplasm that is devoid of discrete cytoplasmic vacuoles. F) Cow 8849, 44 d postpartum. Hepatocytes have numerous discrete cytoplasmic vacuoles after 30 d on KIP.



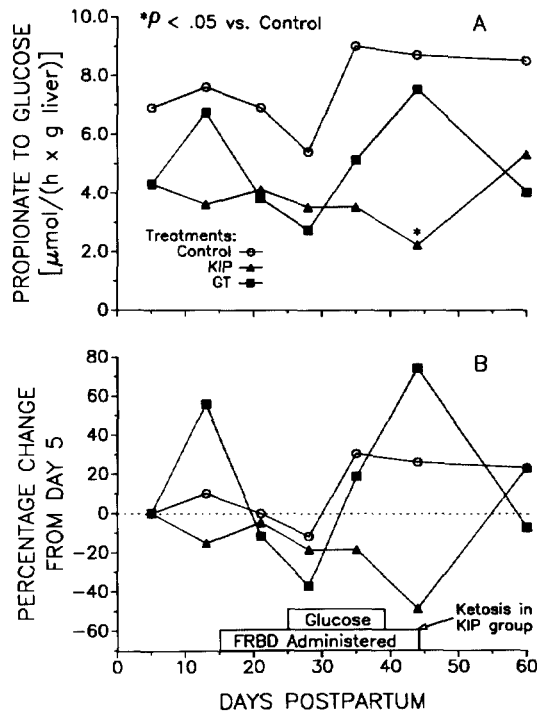


Figure 6. Actual (A) and percentage (B) change from d 5 for conversion of propionate to glucose in slices of liver incubated in media containing 2.0 mM Ca^{++} for control cows, cows given the ketosis induction protocol (KIP), and cows given the glucose treatment (GT). The asterisk refers to differences ($P < .05$) for d 44 for KIP versus control, and FRBD indicates treatment with feed restriction and 1,3-butanediol. Average standard errors for control, KIP, and GT groups were 1.69, 2.00, and 1.81.

Liver biopsies from control cows contained little or no visible lipid after d 14. Figure 5, C and E shows data from a control cow, 8746, biopsied at d 28 and 44, respectively. Figure 5E for control cow 8746 corresponds to the time that cow 8849 (Figure 5F) was clinically ketotic. The control liver sample contained very little fat. The triglyceride contents of the samples shown in Figure 5A to F were .6, 6.0, .8, 8.2, .6, and 9.6% of wet weight, respectively.

In Vitro Incubations

Effects of ketosis induction and intraduodenal glucose treatment on the capacity of liver slices to synthesize glucose from propionate are shown in Figure 6. Previous work (25)

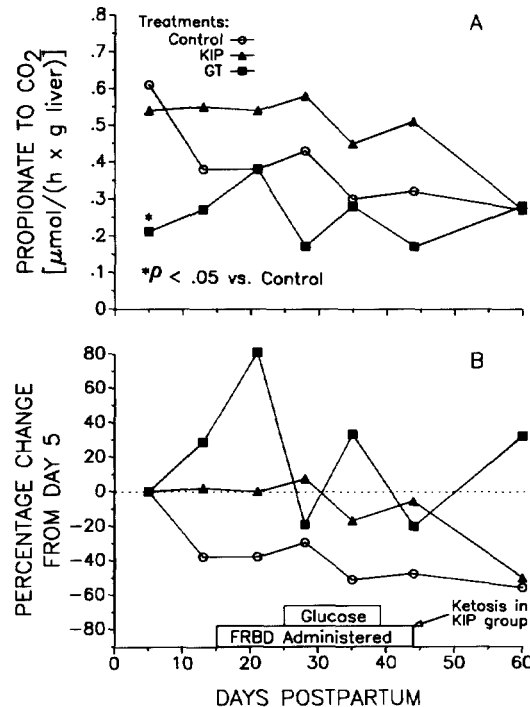


Figure 7. Actual (A) and percentage (B) change from d 5 for oxidation of propionate to CO_2 in slices of liver incubated in media containing 2.0 mM Ca^{++} for control cows, cows given the ketosis induction protocol (KIP), and cows given the glucose treatment (GT). The asterisk for significance at $P < .05$ refers to differences ($P < .05$) for d 5 for GT versus control. Average standard error for control, KIP, and GT groups were .13, .14, and .13.

showed no significant differences for DNA content in cow livers containing from 1 to 11% triglycerides. Therefore, data in Figure 6A are reported as amounts of propionate converted to glucose per gram of liver (wet weight). Data in Figure 6B are expressed as percentage of change from d 5 postpartum and were calculated from Figure 6A. The capacity of liver slices from control cows to synthesize glucose from propionate remained relatively constant for the first 28 d of lactation and then tended to be greater for d 35, 44, and 60, although differences were not significant.

Gluconeogenic capacities for both KIP and GT cows at d 5 and 13 were somewhat (not significantly) lower than for controls (Figure 6A). These nonsignificant differences at d 13 are consistent with differences in blood glu-

cose and liver glycogen for KIP and GT cows at d 13 (Figures 2A and 4A, respectively). Gluconeogenic capacity of liver slices from KIP cows was slightly lower during the first 3 wk of FRBD treatment than for d 5 (Figure 6A), but the decreases were less than 20% (Figure 6B). Control and KIP values were not significantly different during this time. When clinical ketosis was detected at an average of d 44 for KIP, gluconeogenic capacity was decreased even further and was different from controls ($P < .01$). The decrease represented a 50% decline in gluconeogenic capacity from d 5 (Figure 6B), further suggesting that the decrease in gluconeogenic capacity for KIP was associated with the onset of ketosis. A dramatic decrease in gluconeogenic capacity of liver slices when clinical ketosis was detected was observed in an earlier study with a similar FRBD protocol (25). Gluconeogenic capacity quickly returned to early postpartum values during the recovery period for the KIP group (Figure 6).

Gluconeogenic capacity for GT cows was highly variable throughout the study, and no significant differences were noted across time. Gluconeogenic capacity, however, tended to decrease during the first 2 wk of FRBD (Figure 6B) and then to increase during the next 2 wk. Differences in gluconeogenic capacity between KIP and GT cows were significant at d 44 ($P < .04$).

Oxidation of propionate to CO_2 by liver slices is reported in Figure 7 (percentages in Figure 7B were calculated from values in Figure 7A). With the exception of a relatively high mean for controls at d 5, rates of conversion of propionate to CO_2 (Figure 7A) were about 1/10 those for conversion to glucose (Figure 6A). Rates of oxidation of propionate to CO_2 ranged from .2 to .6 $\mu\text{mol/h}$ per gram of liver, and there were no significant differences between control and KIP throughout the study. Oxidation of propionate by liver slices from GT cows was significantly different from control only at d 5 and, subsequently, was unaffected by treatment.

DISCUSSION

Induction of clinical ketosis in the KIP group by using the FRBD protocol confirms previous results from our laboratory (24). The

cows used previously, however, were obese at calving because they were overfed during the dry period. Studies with fasted cows (2, 16) and with spontaneously ketotic cows (3, 29) have shown some profiles of blood and liver metabolites as similar to those noted here at onset of ketosis for the KIP group.

The FRBD protocol created two specific experimental conditions. The first was involuntary feed restriction. The FRBD protocol resulted in about a 9% decrease in energy intake (about 4 Mcal/d decrease) but a 20% decrease in protein intake. The second condition was mild hyperketonemia. Because mild fat accretion has been shown to occur in the liver of some high producing cows when lipids are mobilized for energy in early lactation (26), mild hyperketonemia seemed to be within physiological normalcy for cows in early lactation. Feed restriction alone, however, did not cause ketosis (8), and another study by our group (9) has shown that neither feed restriction nor BD supplementation given separately is sufficient to induce ketosis or the accumulation of hepatic lipid. Therefore, interactions between feed intake relative to production during the first 5 wk of lactation and increased ketones in blood apparently are important for development of ketosis. Cows on all treatments had nearly identical concentrations for all blood and liver measurements taken prepartum. Cows assigned randomly to the KIP treatment before calving, however, tended to have greater milk production, lower plasma glucose concentration, higher liver triglyceride content, and higher plasma NEFA concentration immediately after calving than control or GT cows. The pretreatment differences for metabolites were not statistically significant, yet it is not known whether these pretreatment tendencies affected the responses by either KIP or GT groups.

Two major metabolic differences can be noted between the present versus earlier results (24) in which a similar FRBD protocol was used, except earlier cows were overfed in the dry period. First, prepartum liver glycogen content of our cows was about half that of the earlier cows (24). This likely reflects differences between normal feeding versus overfeeding in the dry period. Second, both control and KIP cows increased liver triglycerides only slightly in early postpartum in the present

study. In the previous study, liver triglycerides increased sixfold after calving (24), suggesting that both a rapid decrease in glycogen and an increase in triglyceride content of liver are prerequisites for susceptibility to ketosis.

Our results concur with the suggestion that fatty liver is a result of negative energy balance and ketonemia (14), because the rapid increase of liver triglycerides in the KIP group developed after ketogenesis began to increase. The photographs in Figure 5 show, however, that fatty infiltration of the liver increased as cows progressed toward clinical ketosis. The actual effect of this fatty infiltration on the onset of clinical ketosis is not known.

Changes in plasma glucose and NEFA concentrations after beginning KIP (Figures 2A and 3A) indicate fat mobilization in response to energy deficit. The subsequent continued increase in BHBA concentration seems to indicate an increased ketogenesis by the liver in order to metabolize NEFA, which also is implied by a gradual decrease in plasma NEFA concentration after d 21 (Figure 3A). Significant increases of plasma acetate (Figure 3C) and liver triglyceride (Figure 4B), however, do not occur until after d 28. Liver fatty acid synthesis was essentially undetectable and was not increased during ketosis (data not shown). These data suggest that the liver utilizes alternate pathways to metabolize acetyl coenzyme A and that lipid transport is impaired (20), indicating the beginning of the disease state. The major increase in plasma acetate for the KIP group is important evidence that this experimental ketosis is similar to spontaneous ketosis, which increases plasma acetate, rather than being simply a starvation ketosis, which results in decreased plasma acetate (22).

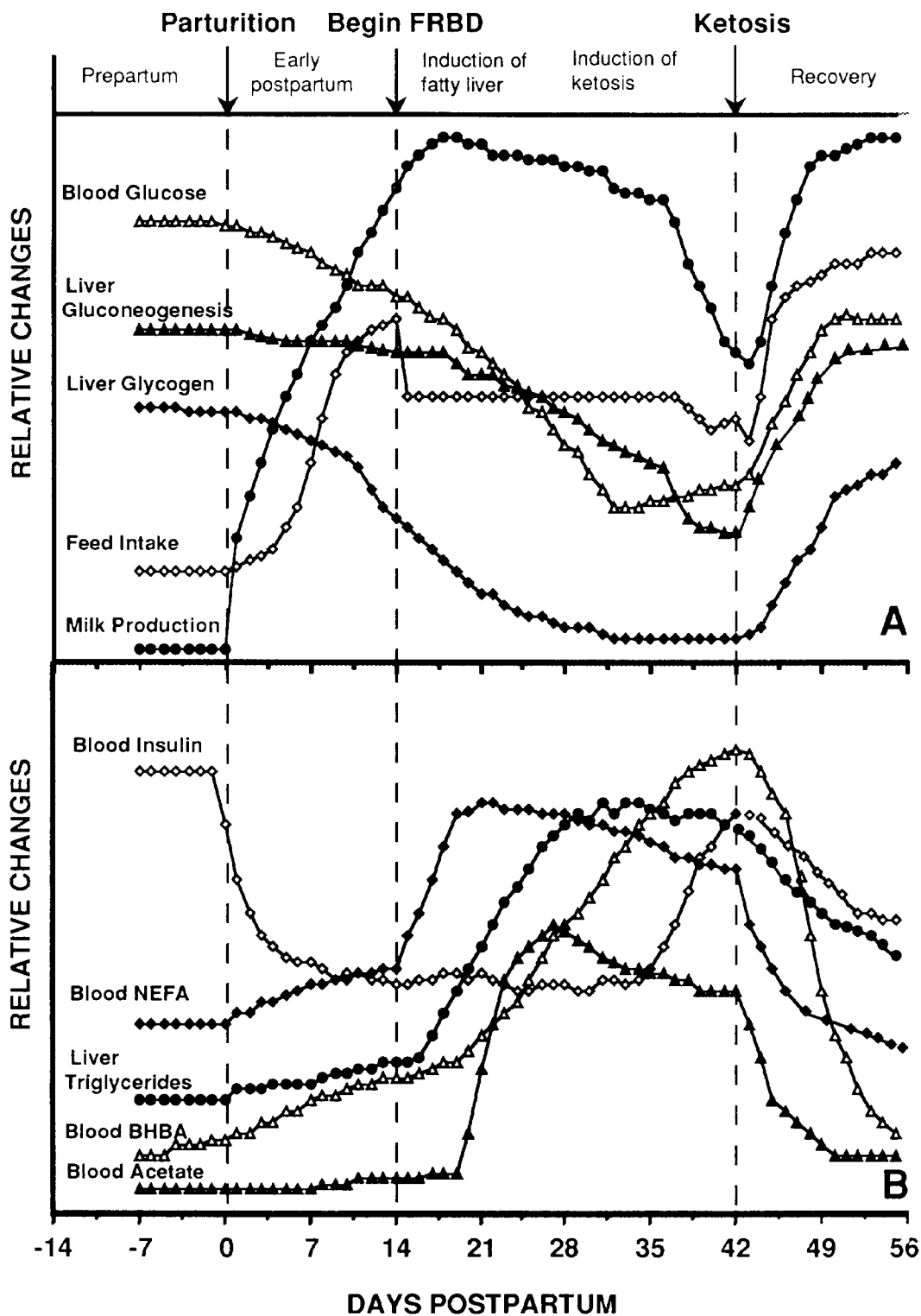
Insulin was measured in the current study to assess one aspect of hormonal regulation of metabolism during development of ketosis. In-

sulin was lower during FRBD for KIP cows corresponding to the time that gluconeogenic substrates were limited because of decreased feed intake. A decrease in insulin concentration during early lactation can promote mobilization of fatty acids from adipose tissue and may indirectly stimulate hepatic gluconeogenesis (6). Contrary to previous work, insulin (Figure 2B) and glucose (Figure 2A) concentrations were increased (nonsignificantly) when KIP cows were diagnosed as clinically ketotic. A decrease in insulin responsiveness in ketotic cows independent of feed intake has been suggested (17). No changes in either insulin or glucose during ketosis were reported when using FRBD (24).

The onset of clinical ketosis at an average of 44 d postpartum was diagnosed after a noticeable decrease in milk production and a further decrease in the already restricted feed intake (Figure 1B). The decrease in milk production at ketosis actually was a continuation of a gradual decrease beginning at the start of FRBD (Figure 1A). Concentrations of metabolites in blood and liver did not change at onset of ketosis from those of 1 and 2 wk earlier. Thus, our data reveal no obvious trigger mechanisms marking the transition from subclinical to clinical ketosis. In vitro measures of hepatic gluconeogenic capacity did show a significant decrease at the onset of ketosis, but it is uncertain whether this was cause or effect.

Changes in blood and liver metabolites seen for KIP cows were absent in GT cows. Explanations of the ability for duodenal glucose infusions to prevent fatty liver and ketosis in GT cows center on two main effects: providing glucose carbon for lactose synthesis in the mammary gland and increasing carbohydrate sufficiency or stimulating regulatory mechanisms to increase hepatic gluconeogenesis (5). The persistency of milk production in the GT

Figure 8. Relative changes (A) of blood glucose, liver gluconeogenesis, liver glycogen, feed intake, and milk production, which decreased before clinical ketosis was induced and (B) of blood insulin, blood NEFA, liver triglycerides, blood BHBA (β -hydroxybutyrate), and blood acetate, which increased before ketosis was induced by FRBD (feed restriction and 1,3-butanediol). Relative rather than absolute changes for individual parameters were generalized, based upon data from the present study and from earlier studies (24, 25) done in our laboratory, and symbols on lines do not represent actual data points. The figure is intended to be a general illustration of the probable scenario as cows progress into fatty liver and clinical ketosis in most on-farm situations.



group (Figure 1A) suggests that duodenal glucose acted to reverse the effects of FRBD. At d 44, difference in milk production between the KIP and GT groups was 9.0 kg/d. Assuming that milk is 5% lactose, 450 g of glucose would be required to synthesize the extra milk from the infused glucose if all other precursors were available. If 65% of the infused glucose was absorbed (10), a maximum of 340 g of glucose/d could be available to the mammary gland. Further, it is generally estimated that, for low producing cows, about 70% of the glucose available is used by the mammary gland for lactose, and infusion of glucose into the abomasum of healthy cows in negative energy balance does not increase milk production (10). The calculated maximum absorbed glucose from infusions in the current study could have provided only about 1.3 Mcal/d of energy. Thus, it seems that milk production of GT cows was maintained by more than just glucose carbon supplied from the intestine.

Because both BD and 484 g of glucose/d were added, metabolizable energy intake of GT cows was about 95% that of control cows. The GT cows showed increases in both NEFA and BHBA after beginning FRBD and before glucose infusion, but the magnitude of the responses was different than for KIP cows (Figure 3, A and B). These confounding data preclude specific conclusions about the action of duodenal glucose. Four major differences in metabolism between the KIP and GT groups were observed, however. First, glucose concentrations in plasma were much greater in GT cows. Second, BHBA concentrations in plasma did not increase in GT cows as in KIP cows, suggesting a decrease in hepatic ketogenesis or an increase in extrahepatic utilization of ketones. Third, liver glycogen of GT cows remained at concentrations observed for control cows, whereas glycogen was nearly depleted in livers of KIP cows. Fourth, there was no increased accretion of triglyceride in liver of GT cows. Body weight loss (Figure 1C), however, changed similarly for the KIP and GT groups; presumably some, but not all, of the loss was associated with a decreased gut fill caused by feed restriction.

The present study is the second to show a gradual induction of ketosis in early lactation cows treated with FRBD. To illustrate the evolving knowledge of the pattern of changes

in metabolic measures during the development of ketosis with FRBD, Figure 8 shows relative changes observed in our study and in earlier studies (24, 25). Two points are useful for evaluation. First, *in vitro* gluconeogenic capacity of liver slices is presented but does not necessarily reflect *in vivo* gluconeogenesis because healthy animals would have excess hepatic gluconeogenic capacity. Second, our animal model of ketosis is experimental, and it is uncertain how it specifically corresponds to on-farm situations.

Figure 8 reflects the experimental conditions, because feed intake was restricted starting at d 14, and blood ketones began to rise. Consequently, plasma NEFA concentrations increased rapidly and peaked about 3 wk before ketosis and remained high. The increase in BHBA in plasma continues until just before ketosis, and plasma acetate was elevated for about 2 wk before clinical ketosis was detected. Most notable in Figure 8B was the increase in liver triglycerides about 2 wk before ketosis detection and the depletion of liver glycogen at the same time. The detection of ketosis was marked by an obvious decrease in milk production, some further decrease in feed intake, and depressed liver gluconeogenesis (Figure 8A). The recovery period is noted by return to normalcy of nearly all measures except liver triglyceride content.

The estimates (Figure 8) from our studies provide insight about the development of clinical ketosis. Although subclinical ketosis is marked by several metabolic aberrations, there is little change when a cow moves from subclinical to clinical ketosis. Although a decrease in hepatic gluconeogenic capacity has been detected at diagnosis of clinical ketosis, it is not possible to conclude whether impaired liver metabolism causes a cow to change from subclinically to clinically ketotic or whether the decreased gluconeogenesis is a result of conditions existing after clinical ketosis occurs.

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